

KALACH, Russian Federation

WMO: 342470

Lat: 50.417N

Lon: 41.050E

Elev: 301

StdP: 14.54

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.7	-6.0	-18.0	2.1	-11.8	-11.9	2.9	-5.2	20.8	30.9	18.0	26.0	2.2	0	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	92.7	66.6	88.9	66.3	85.7	65.4	70.4	84.1	69.0	82.5	67.6	80.8	7.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.8	74.8	64.7	92.6	73.8	63.2	87.8	72.7	34.5	84.0	33.3	82.5	32.2	80.9	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.4	13.4	-19.1	96.6	7.6	3.8	-24.6	99.3	-29.0	101.5	-33.3	103.6	-38.8	106.4		
(5)			-19.4	73.9	7.6	3.3	-24.8	76.2	-29.3	78.2	-33.5	80.0	-39.0	82.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	20.7	21.1	31.6	48.1	60.1	67.0	71.1	68.7	57.7	45.3	34.1	24.6		
	DBStd	20.41	12.50	12.97	9.09	8.11	8.05	6.96	5.83	6.86	7.52	8.24	9.67	11.63		
	HDD50	3907	907	808	571	128	13	0	0	0	19	191	481	789		
	HDD65	7481	1372	1228	1035	508	195	59	12	41	239	610	928	1254		
	CDD50	2440	0	0	1	71	326	511	654	580	248	46	3	0		
	CDD65	537	0	0	0	0	42	119	201	156	19	0	0	0		
	CDH74	6385	0	0	0	25	581	1354	2176	1926	319	4	0	0		
	CDH80	2443	0	0	0	1	163	489	900	813	76	1	0	0		
	WSAvg	5.4	6.1	6.4	6.6	6.1	5.3	4.7	4.1	4.3	4.6	4.9	5.9	6.3		
	PrecAvg	19.8	1.8	1.4	1.2	1.1	1.5	2.1	2.3	1.5	1.9	1.6	1.6	1.7		
(15) Precipitation	PrecMax	27.2	3.5	2.7	2.6	4.4	2.4	4.6	5.7	3.7	5.0	3.6	3.1	3.5		
	PrecMin	13.7	0.6	0.4	0.2	0.2	0.6	0.6	0.4	0.1	0.1	0.2	0.1	0.5		
	PrecStd	3.7	0.8	0.6	0.6	0.8	0.6	1.1	1.3	0.9	1.5	0.9	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	44.8	59.0	77.4	88.9	93.5	97.9	99.5	86.5	72.1	57.4	46.6		
		MCWB	42.0	41.6	46.2	56.5	62.9	65.6	68.4	66.9	62.1	57.5	52.5	44.1		
	2%	DB	38.8	39.9	53.0	71.8	84.6	89.1	92.9	92.7	81.8	67.2	53.3	43.2		
		MCWB	37.0	37.6	44.0	54.7	62.0	66.1	67.6	66.2	60.1	55.6	50.0	41.6		
	5%	DB	36.5	37.2	47.8	67.4	80.7	85.7	88.6	88.3	77.5	62.9	49.8	40.2		
		MCWB	34.8	35.2	42.0	52.4	60.7	65.7	67.5	65.9	59.6	54.6	46.8	38.5		
	10%	DB	34.6	35.4	43.3	63.6	76.5	82.1	84.9	84.2	73.2	58.5	46.3	37.0		
		MCWB	33.3	33.7	38.8	50.9	60.0	65.0	66.3	65.0	58.6	52.4	44.1	35.6		
	0.4%	WB	42.4	42.4	49.3	59.8	66.6	71.9	72.6	71.8	66.8	61.5	53.8	44.6		
		MCDB	44.1	44.8	55.8	72.0	80.1	84.6	86.1	85.8	78.8	68.2	56.1	45.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.5	37.8	45.1	56.6	64.7	69.8	70.7	69.3	64.3	58.3	50.4	41.7		
		MCDB	38.5	39.4	51.6	67.4	78.7	82.1	84.9	83.6	74.0	63.5	53.1	43.1		
	5%	WB	35.0	35.6	41.9	54.5	63.0	67.9	69.4	67.8	62.0	55.4	47.3	38.8		
		MCDB	36.1	36.8	46.8	64.5	76.2	80.1	83.0	82.9	71.1	60.9	49.5	40.0		
	10%	WB	33.4	34.0	39.3	52.2	61.2	66.2	68.1	66.4	59.9	52.7	44.2	35.8		
		MCDB	34.4	35.1	42.9	61.3	73.4	78.2	81.3	80.2	70.2	58.1	46.3	36.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	12.3	13.6	20.5	22.1	21.3	21.7	24.7	22.4	16.3	10.4	9.2		
		MCDBR	6.6	9.1	21.1	27.9	28.6	27.4	28.3	30.7	31.0	24.5	14.9	8.0		
		MCWBR	6.3	7.9	13.3	14.2	11.2	10.2	9.6	9.9	13.5	14.5	11.3	7.5		
		MCDBR	6.4	8.2	18.4	23.2	23.5	22.1	23.0	25.5	22.5	19.8	12.7	7.8		
	5% WB	MCWBR	6.3	7.7	12.6	13.0	10.9	10.0	9.2	9.9	11.8	13.3	10.5	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.352	0.379	0.446	0.410	0.400	0.428	0.438	0.396	0.368	0.345	0.318		
	taud		2.101	2.006	2.077	2.078	2.276	2.334	2.261	2.214	2.308	2.331	2.270	2.169		
	Ebn at Noon		213	236	256	252	269	272	262	252	250	234	206	195		
	Edh at Noon		26	37	42	47	40	38	41	41	33	27	23	21		
(44) All-Sky Solar Radiation	RadAvg		328	645	1062	1386	1840	1964	1912	1704	1157	622	281	189		
	RadStd		68	115	163	134	159	156	134	125	143	93	59	40		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	-259	+191	N/A	

Nomenclature: See separate page